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In[1]:= (*Omnium Information Density Matrix Simulation*)
(*CodeX V3.2.1*)ClearAll["Global`*"];

(*Define the State Vector for Omnium Transition*)
psi = {a, b} / Sqrt[Abs[a]^2 + Abs[b]^2];

(*Create the Density Matrix (rho)*)
rho = Outer[Times, psi, Conjugate[psi]];

(*Display the Matrix*)
Print["Omnium System Density Matrix (rho):"];
MatrixForm[rho]

(*Calculate von Neumann Entropy (Information Content)*)
vonNeumannEntropy = -Tr[rho.MatrixLog[rho]];
Print["System Information Entropy (S): ", vonNeumannEntropy];

(*Visualizing the State on a Bloch Sphere*)
(*This represents the stability of the Immutable Core*)
BlochSphere[rho]

Omnium System Density Matrix (rho):

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Out[5]//MatrixForm=

$$\begin{pmatrix} \frac{a \text{ Conjugate}\left[\frac{a}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}\right]}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}} & \frac{a \text{ Conjugate}\left[\frac{b}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}\right]}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}} \\ \frac{b \text{ Conjugate}\left[\frac{a}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}\right]}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}} & \frac{b \text{ Conjugate}\left[\frac{b}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}\right]}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}} \end{pmatrix}$$

... **MatrixLog**: The function Log is not analytic or defined at 0.

System Information Entropy (S):

$$-\text{Tr}\left[\left\{\left\{\frac{a \text{ Conjugate}\left[\frac{a}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}\right]}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}, \frac{a \text{ Conjugate}\left[\frac{b}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}\right]}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}\right\}, \left\{\frac{b \text{ Conjugate}\left[\frac{a}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}\right]}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}, \frac{b \text{ Conjugate}\left[\frac{b}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}\right]}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}\right\}\right\}\right].$$

$$\text{MatrixLog}\left[\left\{\left\{\frac{a \text{ Conjugate}\left[\frac{a}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}\right]}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}, \frac{a \text{ Conjugate}\left[\frac{b}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}\right]}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}\right\}, \left\{\frac{b \text{ Conjugate}\left[\frac{a}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}\right]}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}, \frac{b \text{ Conjugate}\left[\frac{b}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}\right]}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}\right\}\right\}\right]$$

Out[8]= BlochSphere

$$\left[\left\{\left\{\frac{a \text{ Conjugate}\left[\frac{a}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}\right]}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}, \frac{a \text{ Conjugate}\left[\frac{b}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}\right]}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}\right\}, \left\{\frac{b \text{ Conjugate}\left[\frac{a}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}\right]}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}, \frac{b \text{ Conjugate}\left[\frac{b}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}\right]}{\sqrt{\text{Abs}[a]^2 + \text{Abs}[b]^2}}\right\}\right\}\right]$$